

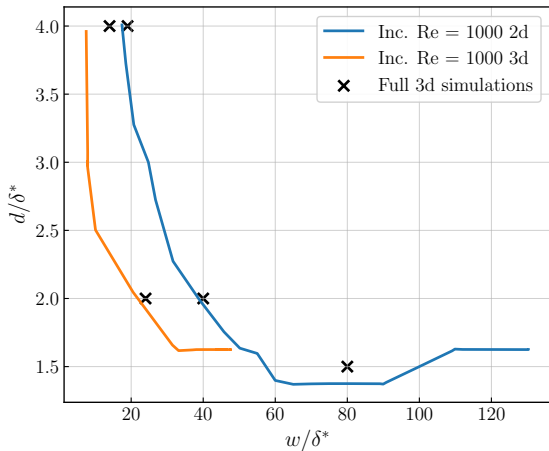
IMPERIAL

3d stuff + 2d TS waves

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Organization Inc work

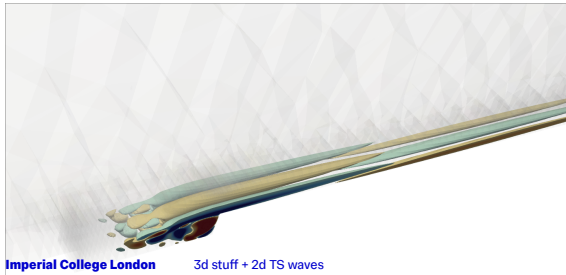
1. 2d hopf bifurcation line as a curve in the plane (w , d).
2. ts mode growth for corroboration of jeff results. good match between “jeff by pass transition boundary” and hopf bifurcation boudnary, even though one simulation is 2d incns and the other is 3d comns. why? this motivates the 3d analysis.
3. start 3d linear stability analysis. we find the that for deep gaps the 3d instability happens before the 2d one. but for shallow gaps the 2d instability happens first.
4. do full 3d (16 complex fourier modes) analysis for different cases:



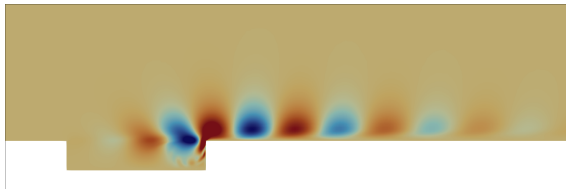
Problems that prevent me to start writing the paper

- Doing 3d linear stability analysis with only one mode, the 3d modes looks more unstable than the 2d. For example for $d = 4$, $w = 19$ (after hopf bifurcation boudnary) the simulations outputs a 3d mode (with $w = 0$) and growth rate $\simeq 4 \cdot 10^{-3}$. 2d LSA give the self-sustained oscilaltion with growth rate $\simeq 2.4 \cdot 10^{-3}$. But in full 3d simulations, after adding small noise to the unstable baseflow, I get first the 2d mode growing with the expected growth rate and expected frequency.

u isovolumes of 3d mode with $k_z = 2\pi/L_z$ with $L_z = 1.25d$ (around the most unstable 3d wavenumber). The mode has $w = 0$.



v heatmap of the 2d mode with at same $d = 4$, $w = 19$.



Parenthesis: SVV consistency

SVV modifies quite a lot the growth rate of the most unstable eigenvector in the system.

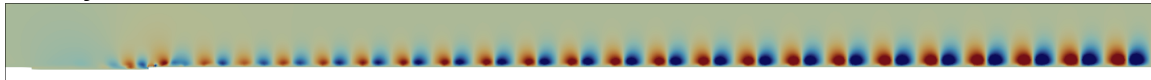
For example for $d = 4, w = 19$:

- $\sigma_{\text{NO SVV}} = 2.4703 \cdot 10^{-3}, \omega_{\text{NO SVV}} = 0.24573$
- $\sigma_{\text{SVV}} = 2.6004 \cdot 10^{-3}, \omega_{\text{SVV}} = 0.24594$
- Relative error in σ is 5.2%, while in ω is 0.08%
- The growth rate with SVV matches the one of 3d dns (because there I have SVV activated as well) up to less than 1% error. And the frequency even better.

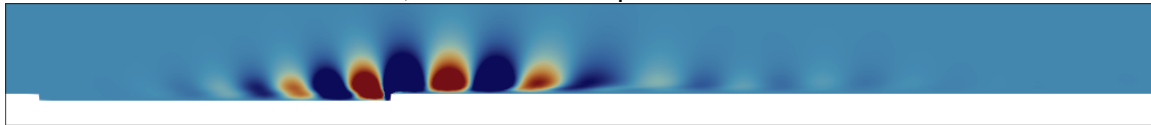
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- For a shallow gap case, like $d = 1.5$, $w = 70$, after DNS finishes the transient state, it stabilizes into a time-periodic signal with the $\omega_{\text{DNS}} \approx 0.116$. However when running 2d LSA, I obtain two unstable modes (at least), each of different frequency: $\omega_1 \approx 0.157$ ($\sigma_1 \approx 0.00368$) and $\omega_2 \approx 0.119$ ($\sigma_2 \approx 0.00237$). So there is an interaction between the two modes that do not produce chaos yet.

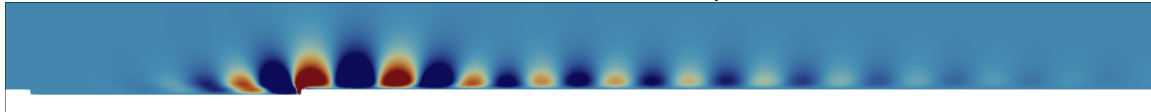
Limit cycle of nonlinear simulation for $d = 1.5$, $w = 70$. v contours.



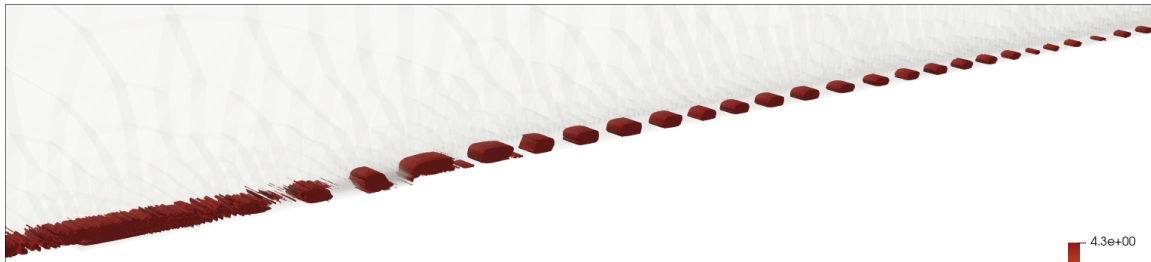
Most unstable 2d mode for $d = 1.5$, $w = 70$. v heatmap.



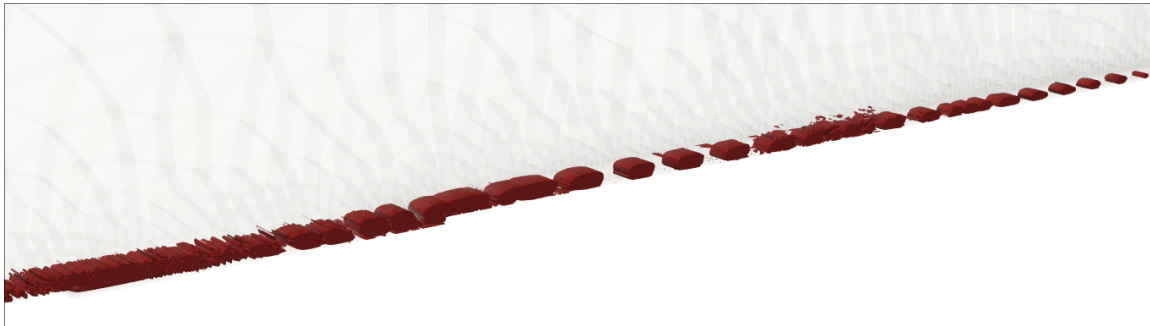
Second most unstable 2d mode for $d = 1.5$, $w = 70$. v heatmap.



Transition process at $d = 1.5$, $w = 80$ (TS mode transition)



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